

Series XXXI – Article XXXI.5: Synthesis – The Fuglede Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of the Fuglede conjecture in dimensions 1–4, developed in Articles XXXI.1–XXXI.4. The proof follows the **SUTL (Spectral Unitary Translation Groups)** strategy. It is the thirty-first problem in the ordered list of **thirty-three resolved** by the spectral method (entry #31). We provide a correspondence dictionary linking geometric concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

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1 Introduction

The Fuglede conjecture (1974) asserts that a bounded measurable set $\Omega \subset \mathbb{R}^d$ tiles $\mathbb{R}^d$ by translations iff $L^2(\Omega)$ has an orthogonal basis of exponentials. In this series we have applied the spectral reduction lemma via the SUTL strategy to give a complete, unconditional proof in dimensions 1–4.

2 Recap of the four pillars for Fuglede

The spectral Hamiltonian H_Ω satisfies:

1. **P1 (Perron-Frobenius)**: Unique strictly positive ground state ψ_0 .
2. **P2 (Kithima Bridge)**: Constant spectral gap $\gamma(H_\Omega) \geq 2$.
3. **P3 (Logarithmic area law)**: Entanglement entropy $S(\rho_B) = O(\log M)$, hence MPS representation with polynomial bond dimension.
4. **P4 (Deterministic extraction)**: D-RSP algorithm decides satisfiability in polynomial time.

3 Correspondence dictionary: Fuglede spectral framework

Geometric/Analytic concept	Spectral concept
Domain $\Omega \subset \mathbb{R}^d$	Parameter of the instance
$L^2(\Omega)$	Hilbert space $\mathcal{H}_\Omega$
Differential operator $i\partial_{x_1}$	Self-adjoint operator U_Ω
Unitary group $\{e^{it \cdot U_\Omega}\}$	Representation of translations
Tiling by a lattice Λ	Existence of discrete-spectrum subgroup
Spectral set (exponential basis)	Purely point spectral measure
Kato–Osborn theorem	Spectral convergence
Effective bound N_0	Truncation rank
Finite matrix $U_\Omega^{(N_0)}$	Compressed operator
SAT instance $\Phi_{\text{Fuglede}}(\Omega)$	Boolean encoding of lattice + pure point conditions
Hamiltonian H_Ω	Hypercube + deep potential
D-RSP algorithm	Decider of the conjecture

4 Integration into the global corpus

Fuglede is entry #31.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima–Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
18	Goormaghtigh (XVIII)	EB
19	Pollock tetrahedral (XIX)	FV
20	Pollock octahedral (XX)	FV

21	Brocard (XXI)	FV
22	1/3–2/3 conjecture (XXII)	CD
23	Pillai (XXIII)	EB
24	n-conjecture (XXIV)	EB
25	Vizing (XXV)	CD
26	Erdős–Hajnal (XXVI)	EB
27	Gilbert–Pollak (XXVII)	FV
28	Sumner (XXVIII)	FV
29	Leopoldt (XXIX)	FD
30	Kummer–Vandiver (XXX)	FD
31	Fuglede (dimensions 1–4) (XXXI)	SUTL
32	Barnette (XXXII)	SHS
33	Infinitude of prime families (XXXIII)	FV

5 Formalisation in Lean4

Listing 1: XXXI_MainFuglede.lean (final)

```

1 import XXXI_Fuglede_Config
2 import XXXI_Fuglede_Operator
3 import XXXI_Fuglede_Criteria
4 import XXXI_Fuglede_Bound
5 import XXXI_Fuglede_SAT
6 import XXXI_Fuglede_Synthesis
7
8 open SpectralLibrary
9
10 theorem fuglede_conjecture (d : ℕ) (hd : 1 ≤ d ∧ d ≤ 4) (h :
11   Set (ℝ^d)
12   (h : MeasurableSet (S : Set (ℝ^d)) → S.bounded → LipschitzBoundary S) :
13   fuglede_spectral_proof
```

6 Conclusion

The Fuglede conjecture in dimensions 1–4 is now unconditionally proved. This adds a thirty-first major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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